

Unit 9 - Data Sets	
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The following B.E.S.T Standards will be covered in this unit:

MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.

MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode and range.

- *Clarification 1: Numerical data is limited to positive rational numbers.*

MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile and the maximum. Use this summary of the data to describe the spread and distribution of the data.

- *Clarification 1: Instruction includes describing range, interquartile range, halves and quarters of the data.*

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers and the range.

- *Clarification 1: Refer to K-12 Mathematics Glossary (Appendix C).*

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.

- *Clarification 1: Instruction includes collecting data and discussing ways to collect truthful data to construct graphical representations.*
- *Clarification 2: Within this benchmark, it is the expectation to use appropriate titles, labels, scales and units when constructing graphical representations.*
- *Clarification 3: Numerical data is limited to positive rational numbers.*

Unit 9, Lesson 1: Types of Data



Benchmark

MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.



Learning Target

☐ I can classify data as categorical or numerical.



9.1.1 Warm-Up: Using Data

At the end of the school year, the school administration will ask students to participate in a survey about their favorite class.

1. What are two possible questions the survey could ask?

2. Makayla thinks it would be helpful to ask students to rank their current classes on a scale of 1-6, with 1 being the students' favorite class and 6 representing their least favorite. Explain how this data could be useful.



9.1.2 Guided Instruction: Types of Data



A **statistical question** is a question that can be answered by collecting data. Often, there will be variability in the data.

Examples and non-examples of statistical questions are shown.

Statistical Questions	NOT Statistical Questions
- How many siblings do the students in my class have? - How tall are the players on the girls' basketball team? - What is the favorite song of the 6th graders at my school?	- What time does school start? - Who is the President of the United States? - What is Avery's favorite color?

- With your partner, discuss the differences between the two types of questions. What do you notice?



Data are values that are collected together for reference or analysis.

- What do you think "variability in the data" means?

- Work together to develop a statistical question. Use your question to gather five data values from your classmates. The answers to your question should all be numeric. Your classmates' responses are the data values.

- Complete the table.

Question				
Data				

- Does the data from your question show a lot of variability or little variability?
- Explain how you determined your description of the variability.

4. Two statistical questions are shown.

Question 1: "How many siblings do the students at my school have?"

Question 2: "How many hours do the teachers at my school spend outdoors on a weekend?"

a. Explain which question you would expect to have more variability.

b. Discuss your choice with your partner.

Data can fall into two different classifications — categorical or numerical.



Categorical data is a type of data that is divided into groups. Categorical data are qualitative.

Numerical data, or quantitative data, is a type of data that is measurable. It is collected in the form of numbers.



5. In the table, classify the type of data described, or provide an example of data with that classification.

Data Description	Examples	Data Classification
m&m colors	red, blue, green, yellow, brown	
number of pets a student has	0, 2, 1, 4, 8, 2, 2, 1	
zip code		
		categorical
		numerical



9.1.3 Exploration: Class Survey

1. Answer the following questions. Write your answers in the table under the number that corresponds to each question.
 - a. How many letters are in your first name?
 - b. How old are you, in years?
 - c. What is your favorite food?
 - d. What is your favorite sport?
 - e. How many languages do you speak fluently?
 - f. What languages do you speak fluently?

Your Responses to Each Question

a.	b.	c.	d.	e.	f.

2. Review the display of the combined class data.
 - a. Which data sets are categorical?
 - b. Which data sets are numerical?



9.1.4 Guided Practice: Classifying Data

1. A city parks department recently collected information about the breeds of dogs visiting a local dog park. They discovered that over 50 breeds of dogs visited the park on Saturday.

Determine if the data is categorical or numerical.
Explain your thinking.

2. At the beginning of the season, the football coach creates a roster with the players' names, ages, grade levels, and uniform numbers.
 - a. Which of the variables are categorical?
 - b. Which of the variables are quantitative?
 - c. For each variable, explain how you determined the type of data.

3. What is the average weight of an elephant, in pounds?

- a. What is your guess? _____
- b. Compare your answer with four other students. What do you notice?

c. Is this categorical or numerical data?

4. Create a statistical question that will result in categorical data being collected.

5. Create a statistical question that will result in quantitative data being collected.



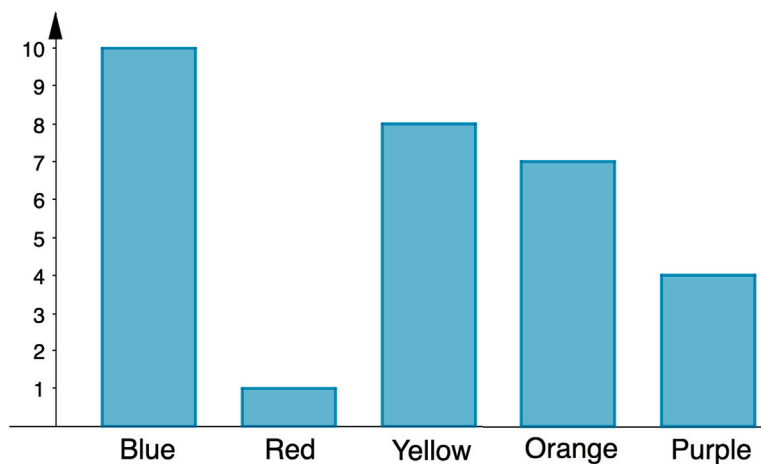
9.1.5 Your Turn!

1. Classify each variable as Categorical (C) or Numerical (N).

- a. Social media followers _____
- b. Color of cars in the faculty parking lot _____
- c. Shoe size _____
- d. Cost of a drink _____
- e. Zip code _____
- f. Type of pets you have _____

2. Will the question, "What is your favorite television or movie streaming service to use?" result in categorical or quantitative data? Explain how you made your choice.

3. Given this bar chart, classify the data as categorical or quantitative.



9.1.6 Wrap-Up: Error Analysis

Bradley's teacher asked him to circle the numerical data in the table below.

Name	Student ID	Age	Hair Color	Grade Level
Rachel	1254866	10	Brown	6
Elijah	2548631	11	Brown	6
Chase	2645852	11	Blonde	6
Ben	2548654	10	Black	6

1. His teacher said that his response was incorrect. Correct his mistake and explain what mistake he made.

Unit 9, Lesson 2: Statistical Questions



Benchmark

MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.



Learning Target

- ☐ I can write statistical questions that generate numerical data.



9.2.1 Warm-Up: Math Talk

In the spaces below, write three statistical questions that you could use today to collect data from your classmates. Be sure at least one of your questions will collect numerical data.

1. _____

2. _____

3. _____



9.2.2 Activity: Gallery Walk

1. With your partner, discuss the following questions.
 - a. How do you know the questions you wrote are statistical questions?
 - b. How can you determine which of the questions will generate numerical data?
2. With your partner, determine the two best questions whose answers provide **numerical data** from the six questions you each created.

Determine which of these two questions each partner will use to collect data about your class.

3. Use your statistical question to collect some numerical data following the directions below. This data will be used in a later lesson, so keep it safe.
 - a. Write your statistical question in the top of the data chart that you chose in the previous problem.
 - b. Place the data chart in a location that is easily accessible by your classmates.
 - c. Follow your teacher's directions to answer each group's question.

My Statistical Question			
Name	Answer	Name	Answer
1.		15.	
2.		16.	
3.		17.	
4.		18.	
5.		19.	
6.		20.	
7.		21.	
8.		22.	
9.		23.	
10.		24.	
11.		25.	
12.		26.	
13.		27.	
14.		28.	



9.2.3 Your Turn!

1. Give two examples and two counterexamples of a statistical question.

Examples of Statistical Questions	Counterexamples that are NOT Statistical Questions

2. Write the definition of a statistical question in your own words.
3. The question, "How tall is the tallest man in the world, in inches?" is not statistical. Use your definition of a statistical question to justify why it is not a statistical question.
4. Decide if the question, "How many miles do you live from the school campus?" is statistical or non-statistical. Explain your decision.



9.2.4 Wrap-Up: Reflection

1. What is the most difficult part about writing statistical questions?
2. How can it be determined whether or not a statistical question can be answered with numerical data?
3. What was the most surprising response to your statistical question? Why?

Unit 9, Lesson 3: Finding and Interpreting the Mean, Median, Mode, and Range



Benchmark

MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.



Learning Targets

- ☐ I can find the mean, median, mode, and range of a given data set.
- ☐ I can interpret the mean, median, mode, and range of a given data set.



9.3.1 Warm-Up: Stripes

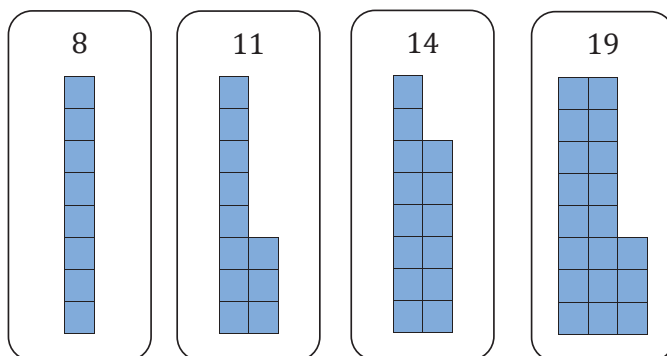
A picture of four sets of 16 cubes is shown.



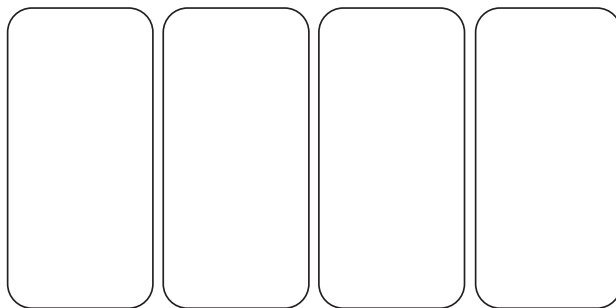
1. Which set of cubes has a set of stripes that covers the most area?
2. Explain how you determined your answer.
3. Which set of cubes has a set of stripes that covers the least area?
4. Explain how you determined your answer.



9.3.2 Exploration: Sharing Squares



1. Estimate the number of squares that will be in each group if they were distributed evenly between the four groups.
2. Use the diagram to redistribute the squares so that each of the four groups has the same number of squares.



3. Compare diagrams with your partner. Discuss how you each determined the number of squares that belong in each group. Summarize your discussion.



9.3.3 Collaboration: Mean of Data

By redistributing the squares in the diagram, the **mean** of the data set was found.



The **mean** is the arithmetic average of a set of numbers. It is a measure of central tendency.

Data in a numerical set is summarized using different measures that are interpreted to describe the data. One such measure used to describe quantitative data sets is the mean.

To calculate the mean, add all the values in the data set and then divide by the total number of data values.

1. A marine biologist is studying turtles and wants to calculate the average length of time a Red-Eared Slider, a popular breed of turtle, can remain underwater. The time, in minutes, that ten Red-Eared Sliders remain underwater are 75, 300, 225, 400, 250, 195, 320, 45, 165, 410.
 - a. To calculate the mean time spent underwater, start by finding the _____ of the data values.
 - b. Then divide the sum by 10, because there are ten turtles.
 - c. In this sample, Red-Eared Slider turtles remain underwater for an average of _____ minutes.



Measures of center are numerical values used to describe the overall clustering of data in a set, or the overall central value of a set of data. The three most common measures of central tendency are the mean, median, and mode.



9.3.4 Guided Instruction: Mean, Median and Mode

- For each definition, fill in the blank with one of the words from the bank.

median

mode

mean

range



The _____ is the arithmetic average of a set of numbers. It is a measure of central tendency.



The _____ is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.



The _____ is the value found most often in a set of numbers. There may be no, one, or more than one _____ in the data set. It is a measure of central tendency.



The _____ is the middle of an ordered list of the values. If the list has an odd number of values, it is the middle value of that list. If the list has an even number of values, it is the average of the two middle values. It is a measure of central tendency.

The number of pets ten students in Tyler's class have is reflected in the table.

Number of Pets									
1	0	2	1	10	0	2	0	8	12

2. Describe steps to find the median.
3. The median number of pets for the ten students is _____.
4. The mode of the data set is _____.
5. Write an explanation of what the mode of the data set means for the context.
6. Describe the steps needed to find the average number of pets owned by the ten students.
7. The mean number of pets for the ten students is _____.
8. Complete the statement about the mean of the data set.
"Each student in Tyler's class has about _____ pets."





9.3.5 Try It: Mean, Median, and Mode

1. Nine adults boarding a cruise ship at Port Canaveral in Florida were randomly surveyed on the total number of vacations they have taken in the last five years. Their totals are summarized below.

Number of Vacations Taken								
5	4	3	9	1	4	0	6	3

- a. What does the value of the median represent in the context?
- b. What is the median number of vacations taken by the adults surveyed?
- c. Determine the mean of the data. Then, interpret the mean in terms of the context by writing a sentence to summarize. Round to the hundredths.



9.3.6 Exploration: Understanding Mode and Range

Work with your partner to complete the following.

1. The

- ☐ mean
- ☐ median
- ☐ mode
- ☐ range

is the value occurring most often in

a set of data, which can be used to describe trends in data.

2. While shopping over the last month, Shep collected the price for a jar of pickles from eight stores he visited.

Price for a Jar of Pickles			
Store A	\$1.66	Store E	\$1.99
Store B	\$2.99	Store F	\$1.99
Store C	\$1.49	Store G	\$1.50
Store D	\$1.99	Store H	\$1.99

- a. What does a jar of pickles cost at most stores?

- b. This value is the

- ☐ median of the data set.
- ☐ mode of the data set.

- c. What is the most Shep will spend on a jar of pickles?

d. What is the least Shep will spend on a jar of pickles?

e. The range of a data set describes how spread out the data is. Determine the range in price for a jar of pickles.

3. Shep also began collecting the price of his favorite cereal to determine where he would get the best price.

Price for a Box of Cereal			
Store A	\$3.79	Store E	\$3.99
Store B	\$3.89	Store F	\$3.89
Store C	\$3.99	Store G	\$4.29
Store D	\$3.89	Store H	\$4.59

a. The mode price of his favorite box of cereal is _____.

b. What is the most Shep will spend on a box of cereal?

c. What is the least Shep will spend on a box of cereal?

d. The range for the price of a box of cereal is _____.

e. The range describes the

☐ center
☐ variability

 of the data set.

4. With your partner, discuss whether the price for pickles or the price for cereal has more variability in pricing. Summarize the discussion.



Measures of variability are numerical values that measure how much a data set varies from a central value.



The range is a **measure of variability**. By comparing the ranges for two or more data sets, the data set that has more variability can be determined by finding which has a larger _____.



9.3.7 Wrap-Up: Reflection

1. Read each of the “I can” statements below. Check the boxes that best explain your understanding.
 - ☐ I can find and interpret the median, mode, and range of a given data set.
 - ☐ I understand how to find the mean, median, and mode of a data set.
 - ☐ I need support on how to find the mean, median, and mode of a data set.
 - ☐ I understand how to find the range of a data set.
 - ☐ I need support on how to find the range of a data set.
 - ☐ I understand how to interpret measures of central tendency.
 - ☐ I need support on how to interpret measures of central tendency.

Unit 9, Lesson 4: Interpreting Box Plots



Benchmark

MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data.



Learning Targets

- ☐ I can determine the minimum, lower quartile, median, upper quartile, and maximum from a box plot.
- ☐ I can determine the interquartile range from a box plot.
- ☐ I can describe the spread and distribution of data from a box plot.



9.4.1 Warm-Up: Which One Doesn't Belong?

Examine the four numbers.

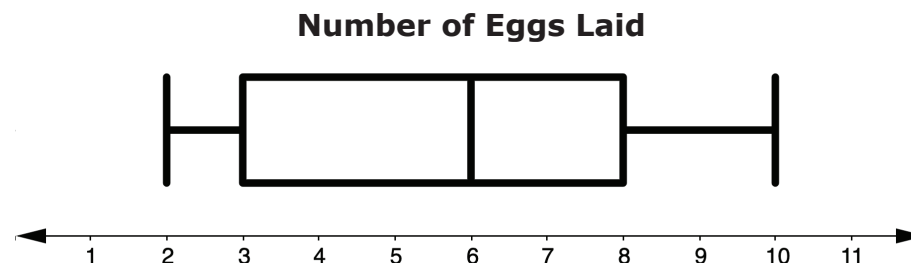
33%	$\frac{1}{3}$
$\frac{5}{3}$	$0.\bar{6}$

1. Which of these numbers does not belong? Explain your thinking.



9.4.2 Exploration: Exploring Box Plots

1. A farmer recorded the number of eggs her hens laid in one week. The data set was the total number of eggs each of her 53 hens laid during the week. Below is a box plot representing the number of eggs laid by each hen during the week.

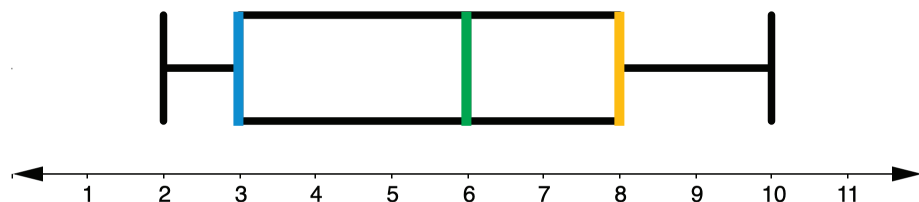


With your partner, use the box plot to complete the statements.

- a. The smallest number of eggs laid by the hens is _____.
- b. The largest number of eggs laid by the hens is _____.
- c. Find another pair of partners and compare the smallest and largest number of eggs laid. Discuss how you determined the values from the box plot. If your values are different, come to a consensus on the smallest and largest number of eggs laid by the hen, and circle these values on the box plot.

- d. Label the range for the number of eggs laid on the box plot.
- e. With your partner, conjecture the median number of eggs laid. Label this number as "median" on the box plot.
- f. The first quartile is highlighted in blue, the second quartile is highlighted in green, and the third quartile is highlighted in yellow.

Number of Eggs Laid



If the second quartile is the median, M , write a conjecture with your partner about how the first and third quartiles are each determined.

2. Work with your partner to complete the statement.

A **box plot** divides data into _____ equal parts that each contain roughly 25% of data from a given data set.

3. Explain why you chose the value you used to complete the statement.

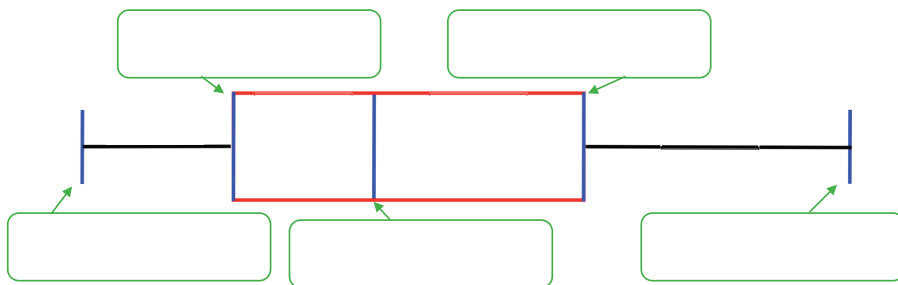


9.4.3 Guided Instruction: Interpreting Box Plots



A **box plot** is a plot displaying the spread or distribution of a data set using a five-number summary: the minimum, lower quartile (first quartile), median, upper quartile (third quartile), and maximum. It is also called a box-and-whisker plot.

1. Label the maximum, minimum, median, first quartile, and third quartile on the box plot.



2. Complete the statement.

The **first quartile (Q_1)** is the middle value of the data

that are

- ☐ less than
- ☐ greater than
- ☐ equal to

the median.

The **third quartile (Q_3)** is the middle value of the data

that are

- ☐ less than
- ☐ greater than
- ☐ equal to

the median.



For a data set with median, M , the first **quartile** is the median of the data values less than M , and the third **quartile** is the median of the data values greater than M . The second **quartile** is the median, M .

3. Complete the statement.

The measure of central tendency used to create a box plot is the
☐ mean,
☐ median,
☐ mode,
 which divides the data set in half.

A number line often accompanies a box plot and is used to determine the values associated with each part of the plot.

Another measure of variability like **range**, the **interquartile range (IQR)**, can be found by determining the difference in the third quartile from the first quartile.

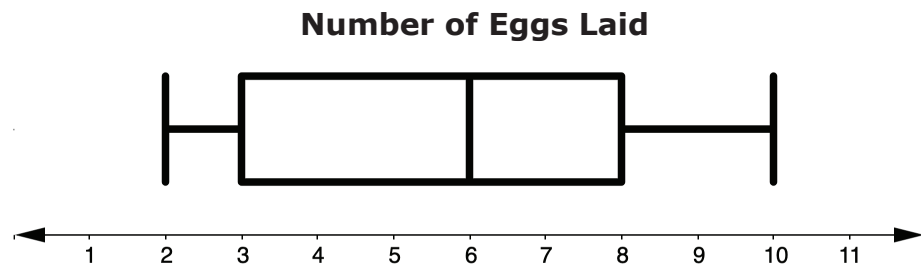


A measure of variation in a set of numerical data, the **interquartile range** is the distance between the first and third quartiles of the data set.



$$IQR = (Q_3) - (Q_1)$$

4. Revisit the box plot of the eggs laid by a farmer's hens.



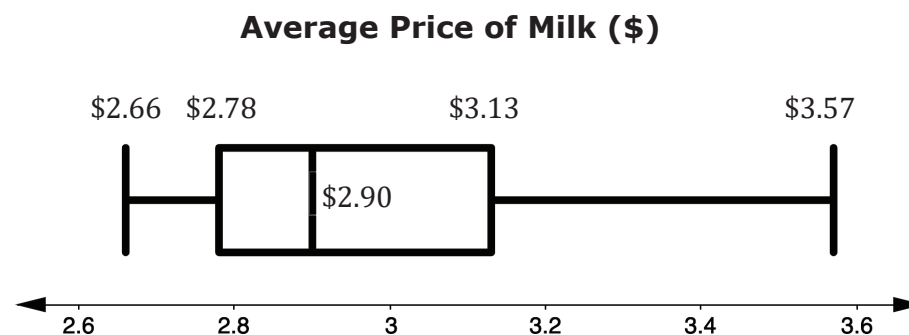
- a. What is the interquartile range (IQR) for the box plot of the context?

- b. Label the IQR on the box plot.



9.4.4 Guided Practice: Interpreting Box Plots

1. The average price of milk, in dollars (\$), for each month from April 2000 to November 2006 was used to create the box plot shown.

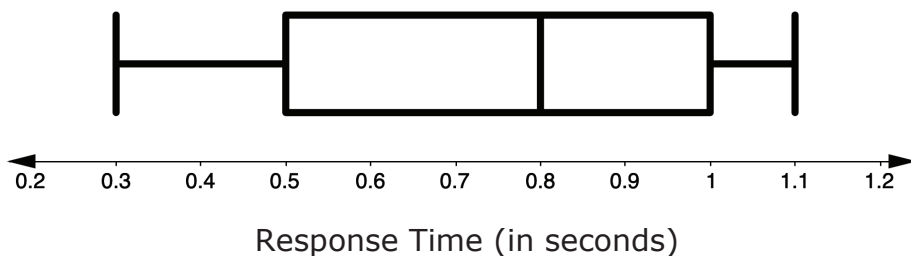


- a. What is the range of the data?
- b. Determine the interquartile range.
- c. Does the average cost of milk from April 2000 to November 2006 vary a lot or a little? Explain your reasoning.



9.4.5 Try It: Interpreting Box Plots

1. The box plot displays data on the response times of 100 mice when seeing a flash of light.

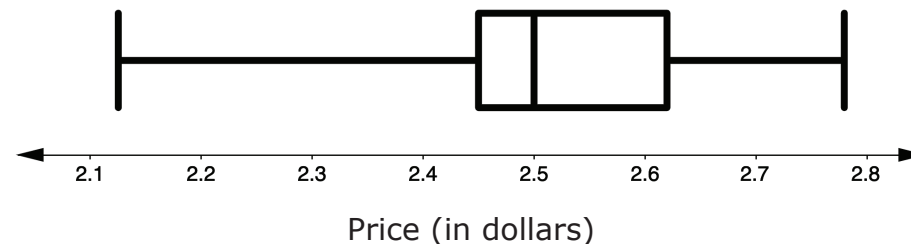


Complete the statements.

- The fastest response time is _____ seconds.
 - The slowest response time is _____ seconds.
 - The middle response time is _____ seconds.
 - The lower quartile is _____ seconds.
 - The upper quartile is _____ seconds.
 - The range is _____ seconds.
 - The interquartile range is _____ seconds.
2. With your partner, use the values to write a description of the spread and distribution of response times.

3. The U.S. Energy Information Administration reports weekly average gas prices for the United States. The box plot represents the weekly average gas prices in 2019.

2019 Weekly Gas Prices



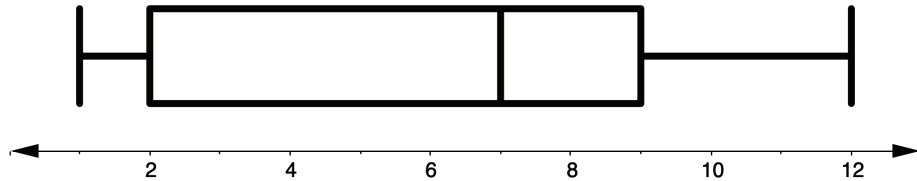
What percentage of weeks had an average gas price of at least \$2.50?



9.4.6 Your Turn!

1. Mrs. Henry's 3rd period English class took a spelling test with 12 words on it. A box plot was created from the number of questions students got correct on the test.

Number of Correct Questions

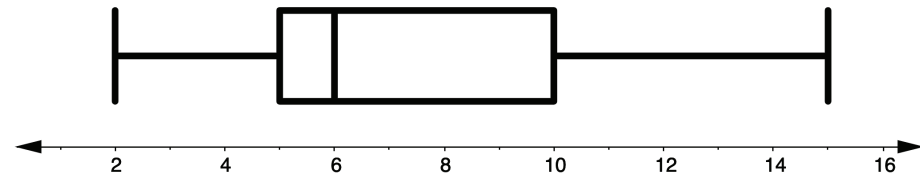


Complete the statements.

- The lowest score is _____.
- The highest score is _____.
- The middle score is _____.
- The lower quartile is _____.
- The upper quartile is _____.
- The range is _____.
- The interquartile range is _____.
- Write a description of the spread and distribution of the scores.

2. The librarian at Franklin Middle School surveyed students when they returned from summer break on how many books they had read. She created a box plot to represent the results of her survey.

Number of Books Read



What percentage of students read between:

- 5 and 10 books over the summer? _____
- 10 and 15 books over the summer? _____
- 5 and 6 books over the summer? _____
- 2 and 10 books over the summer? _____



9.4.7 Wrap-Up: Cool Down

Jamal missed class today and he needs help with what he missed.

1. Write a note to Jamal letting him know what a box plot is and what it is used for. Draw a picture or diagram that may help him better understand.



Unit 9, Lesson 5: Describing and Interpreting Histograms – Part 1



Benchmark

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.



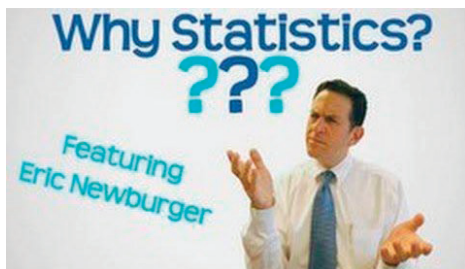
Learning Target

- ☐ I can qualitatively describe and interpret a histogram within a real-world context.



9.5.1 Warm-Up: Why Statistics?

Eric Newburger relates the need for studying statistics to the need for using a microscope. He discusses how a microscope is used to study small things. It's also needed to study things like food poisoning at a larger level to aid in understanding diseases and how viruses spread.



In Mr. Newburger's work, he explains you can take a big thing and shrink it down small enough to study and to understand. That's the study of statistics. And when you want to understand something that is too small, you can use a tool (like the microscope), which is "why statistics."

1. List two reasons why statistics are important.



9.5.2 Guided Instruction: Histograms

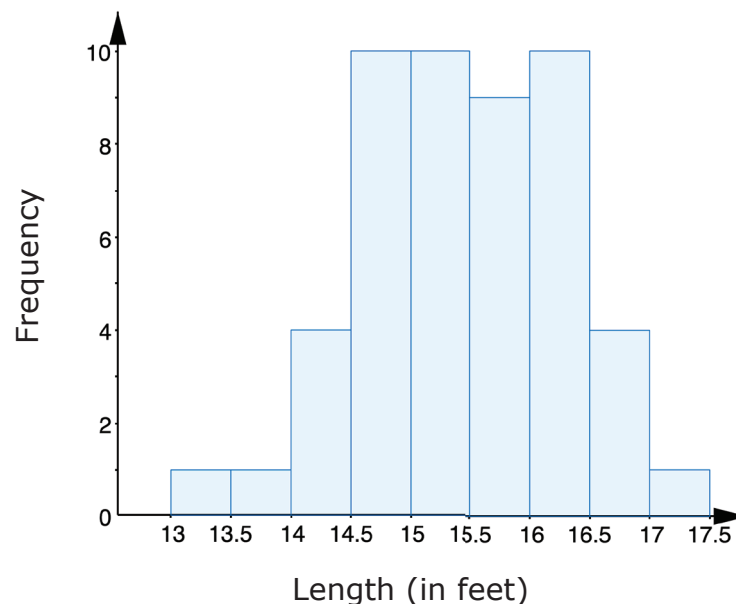
Numerical data can be displayed graphically in a variety of ways. One type of graph is a histogram.



A **histogram** is a visual display of numerical data using bars along a number line with no spaces between the bars. The height of each bar represents either the frequency or relative frequency of data within that interval.

1. The histogram summarizes the lengths, in feet (ft), of a group of adult female sharks.

Shark Length



- a. With your partner, discuss the labels for the vertical and horizontal axes.

b. How many sharks were between 15.5 ft and 16 ft?

c. How many total sharks were measured to create the histogram?

2. Each bar of a histogram includes the left-end value, but not the right-end value. For example, the first bar includes sharks that are 13 ft and 13.25 ft, but not 13.5 ft.

a. Discuss with your partner whether or not you can determine the exact lengths of each shark measured from the histogram. Summarize your discussion.

b. Discuss with your partner your observations about the width of each bar.

3. If male sharks were added to the sample, hypothesize what would happen to the spread of the data.

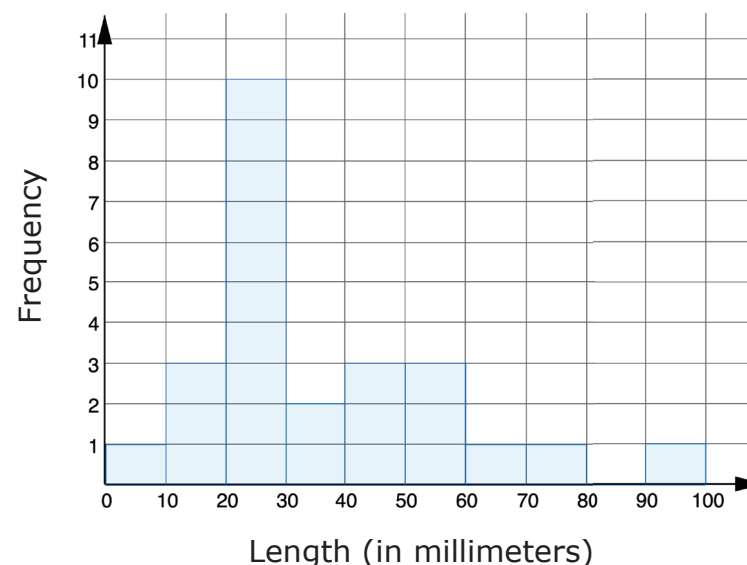


9.5.3 Exploration: Histograms

An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer recorded the earthworms' lengths, in millimeters, in a table and then created a histogram to display the data.

Earthworm Lengths in Millimeters (mm)									
6	11	18	19	20	23	23	25	25	26
27	27	28	29	32	33	41	42	48	52
54	59	60	77	93					



1. How many earthworms were measured in the data?

2. The number of earthworms measured can be

determined from

- ☐ the data set.
- ☐ the histogram.
- ☐ the data set and the histogram.

3. The length of the shortest earthworm can be

determined from

- ☐ the data set.
- ☐ the histogram.
- ☐ the data set and the histogram.

4. Place a "*" above the bar that contains the value for the shortest earthworm.

5. Determine the length of the longest earthworm. Then, place a "+" above the bar that contains the value for the longest earthworm.

6. The mean length of the earthworms can be determined

from

- ☐ the data set.
- ☐ the histogram.
- ☐ the data set and the histogram.

7. What is the mean length of the earthworms?

8. Half of the earthworms are longer than _____ millimeters.

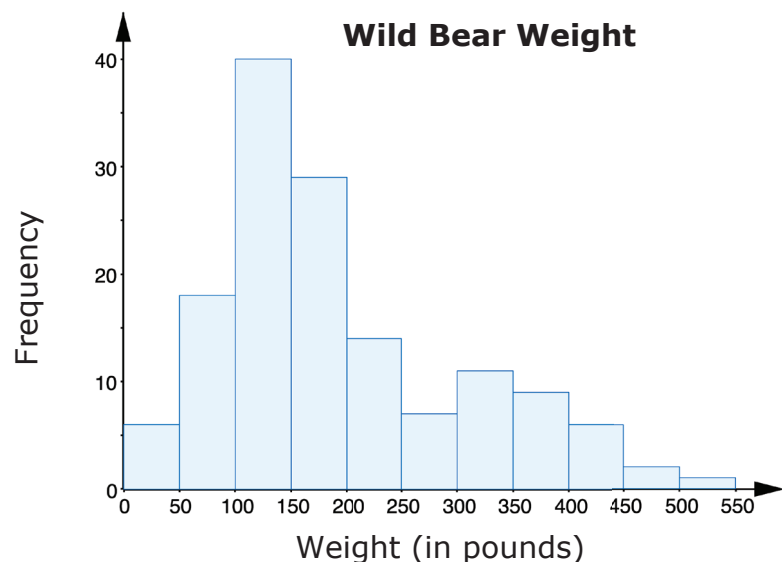
9. Mark an "X" above the bar that contains the mean and mark an "M" above the bar that contains the median.

10. Discuss with your partner how the histogram would change if a 100-millimeter long earthworm was added to the data set. Summarize your discussion.



9.5.4 Try It: Analyzing Histograms

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



- a. Could this sample contain a bear weighing 250 lbs? Explain.
- b. Mark a "*" above that bar that would contain a 250-lb bear.

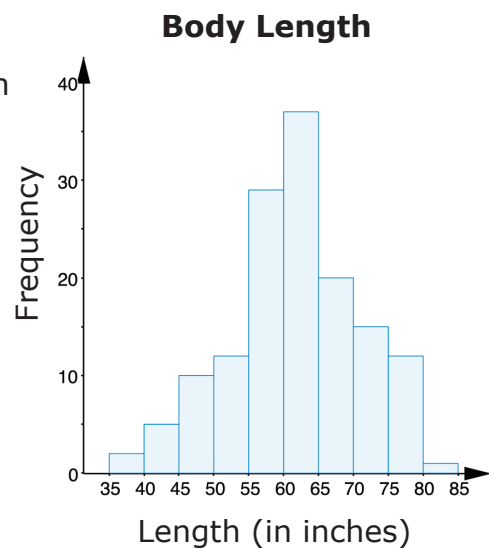
- c. Based on the graph, what can be concluded about the weight of the heaviest bear?
- d. What is a typical weight for the bears in this graph? Include units.
- e. Do more than half of the bears in the study weigh less than 250 lbs?



9.5.5 Your Turn!

1. The histogram summarizes the data on the body lengths of the same 143 wild bears.

- a. Does this sample contain a bear that is 85 inches long? Explain your reasoning.

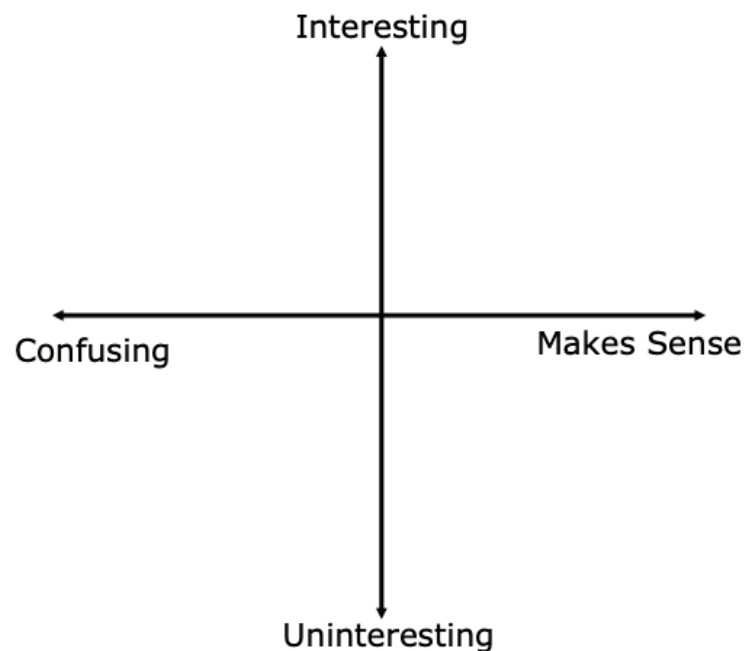


- b. What is a typical length for the bears in this sample? Include units.
- c. Is the spread larger for the weight of bears or the length of bears?



9.5.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate your level of understanding of today's learning target: "I can qualitatively describe and interpret a histogram within a real-world context."



Unit 9, Lesson 6: Describing and Interpreting Histograms – Part 2



Benchmark

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.



Learning Targets

- ☐ I can describe the shape of a histogram.
- ☐ I can describe the spread of a data set given a histogram.
- ☐ I can describe gaps, outliers, and clusters in a histogram.



9.6.1 Warm-Up: Math Fail

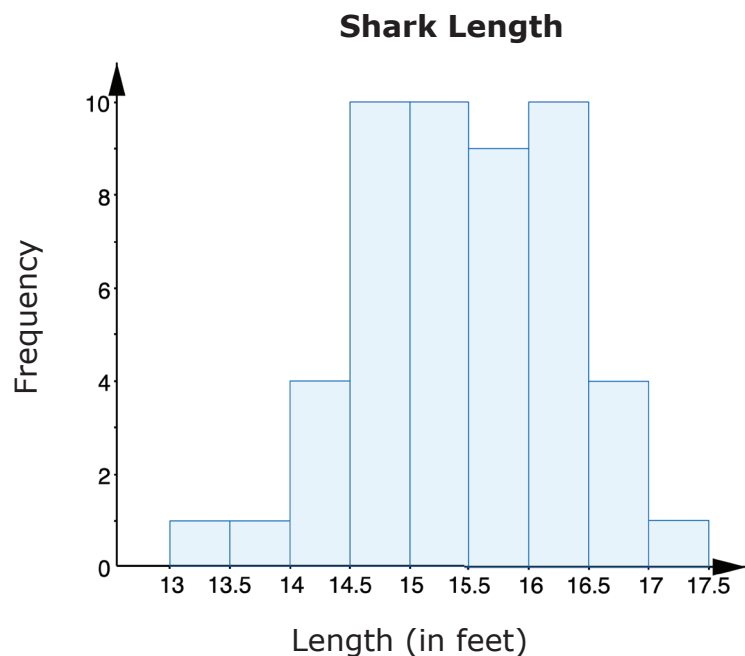
1. Discuss with your partner what you notice about this picture.





9.6.2 Guided Instruction: Shape of Distributions

1. Consider the histogram of the lengths of female sharks.

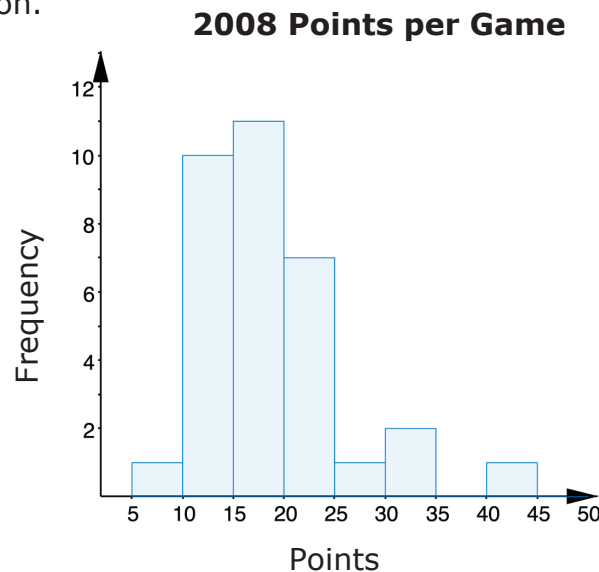


- a. Sketch an approximate line of symmetry for the histogram.
- b. If the histogram is folded along the line of symmetry, will each half line up perfectly?

- c. Do you think all histograms will be symmetrical?

The shape of the shark lengths histogram can be described as **symmetric**, but histograms can also have other shapes.

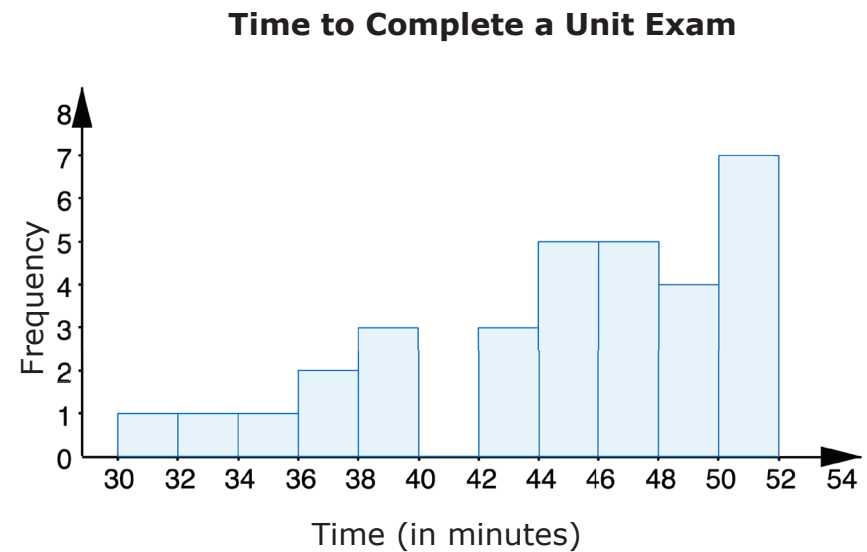
2. The histogram displays the number of points per game scored by a college basketball player during the 2008 season.



Discuss with your partner what you notice about the shape of the distribution of points scored per game.

A **right-skewed** graph has a longer trail of data to the right and the peak of the data on the left.

3. The time to complete a unit exam for 32 students is shown on the histogram.



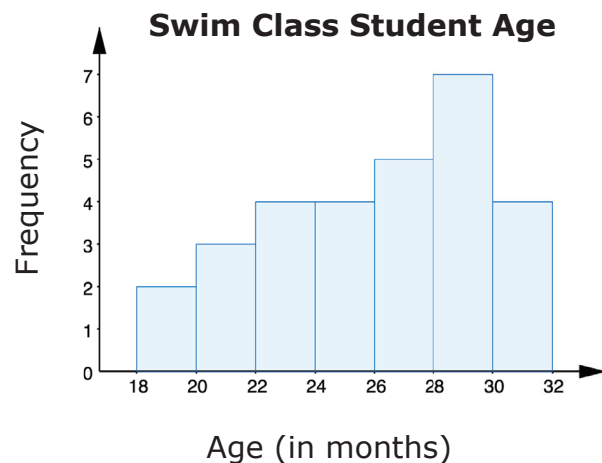
Write an observation about the shape of the distribution of time spent to complete the exam.

A **left-skewed** graph has a longer trail of data to the left and the peak of the data on the right.



9.6.3 Guided Practice: Describing Shape of Distributions

1. This histogram shows the ages of students in a swimming class.

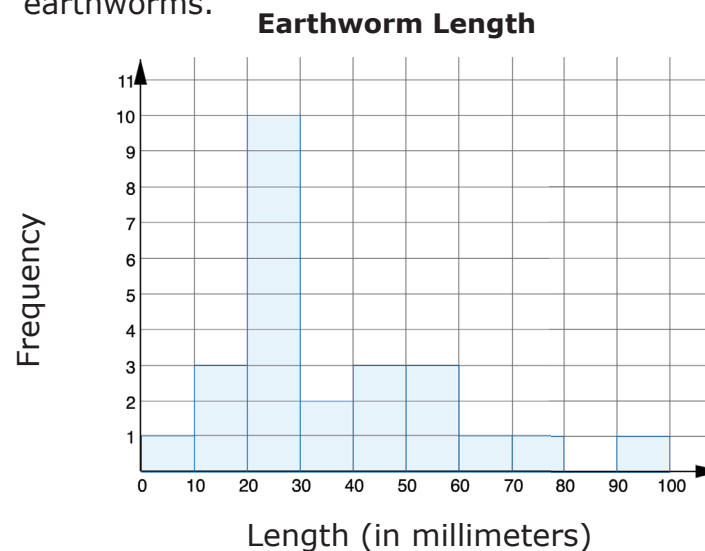


The shape of the distribution of ages can be best

described as

- ☐ symmetric.
- ☐ right skewed.
- ☐ left skewed.

2. The histogram shows the length, in millimeters, of 25 earthworms.

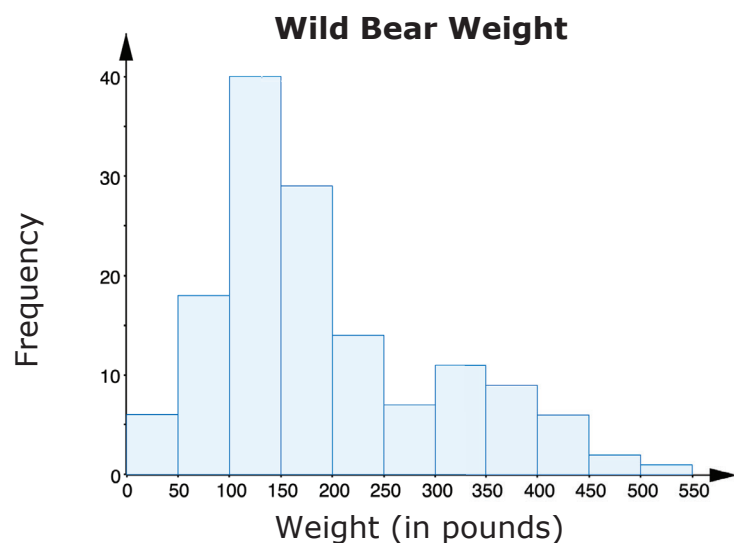


The shape of the distribution of the earthworms'

lengths can be best described as

- ☐ symmetric.
- ☐ right skewed.
- ☐ left skewed.

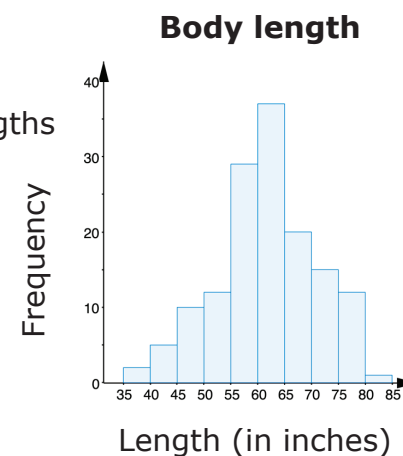
3. In one study of wild bears, researchers measured the weights, in pounds, of 143 wild bears that ranged in age from newborn to 15 years old. The data were used to make this histogram.



Describe the shape of the graph given the context.

4. The histogram summarizes the data on the body lengths of the same 143 wild bears.

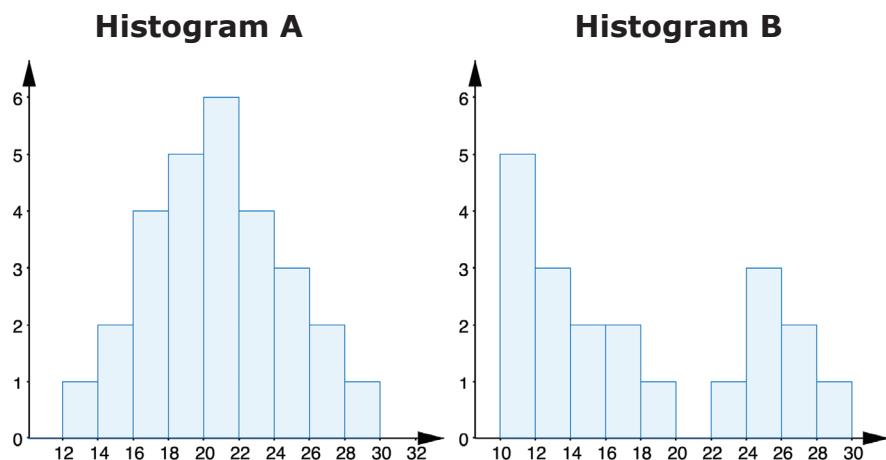
Describe the shape of the distribution of the lengths of bears.





9.6.4 Guided Instruction: Features of Histograms

Below are two histograms with different shapes and features.



1. Histogram A is ☐ symmetrical and has a peak near _____.
☐ not symmetrical

2. Histogram B is ☐ symmetrical and has
☐ not symmetrical
- ☐ one peak.
☐ more than one peak.



A **cluster** is formed when data points are in a close group or numerical data have similar values on a number line.

3. ☐ Histogram A has two clusters.
☐ Histogram B

A **gap** shows a location with no data values. On a histogram, a space between the bins indicates a gap, or a frequency of zero for that bin.

4. ☐ Histogram A has a gap between ____ and ____.
☐ Histogram B



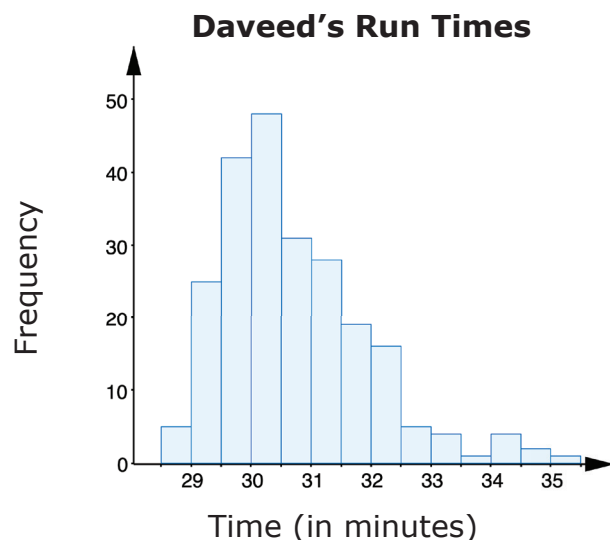
An **outlier** is a value that is much higher or much lower than the other values in a data set.

5. If the data value ☐ 12
☐ 24
☐ 40 were added to Histogram A, it would be considered an outlier.



9.6.5 Collaboration: Features of Histograms

1. Daveed has been tracking the time, in minutes, it takes him to run 4 miles for the last 10 years. A histogram of Daveed's run times is shown.



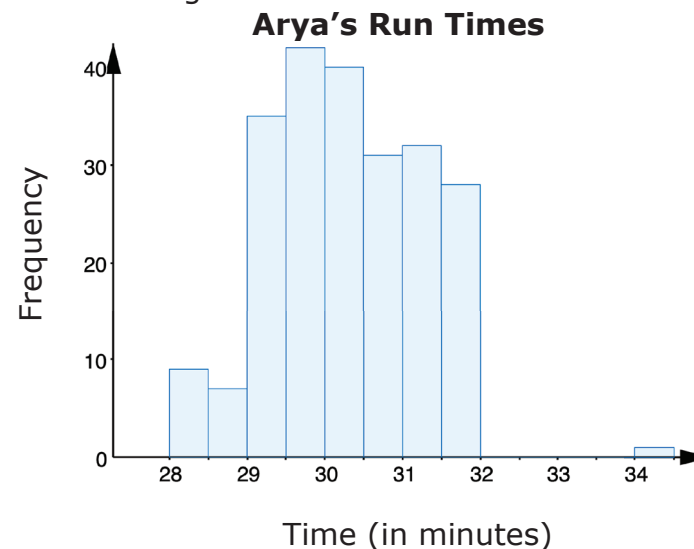
- a. The shape of run times can be best described

as

- ☐ symmetrical.
- ☐ right skewed.
- ☐ left skewed.

- b. Daveed's typical run time was about _____ minutes. Most of his run times were between _____ to _____ minutes. Not many of his runs were ☐ shorter ☐ longer than 32.5 minutes.

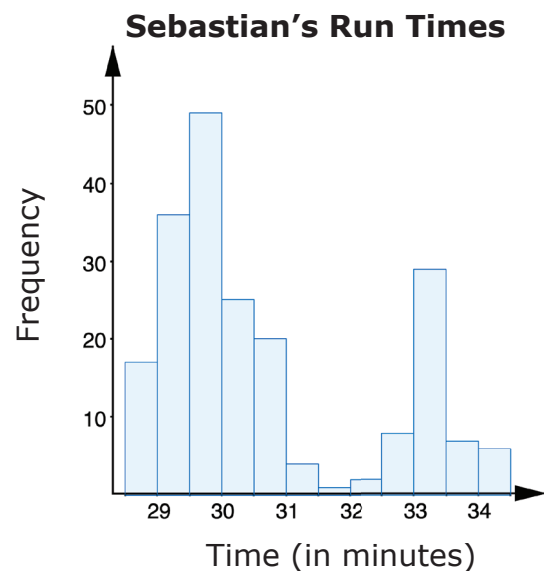
2. Arya used the data of her 4-mile run times to create a different histogram.



- a. Discuss with your partner what you notice about the interval widths. Determine if there are any unusual features including clusters, gaps, or outliers. Write a detailed description of your discussion.

- b. At what interval in the histogram of Arya's times is there an outlier?
- c. Would this same data value also be considered an outlier in the histogram of Daveed's times?

3. Sebastian used the data of his 4-mile run times to create a third histogram.



- a. Which feature is depicted in Sebastian's histogram?

- b. Describe the spread and shape of the distribution of Sebastian's run times.

4. Different data will change the intervals for the horizontal axis and will affect the histogram and its features: symmetry, skewness, gaps, clusters, outliers, and the range. With your partner, review all three run-time scenarios and discuss the differences between the histograms.



- Sixth Graders**

Frequency

Number of Text Messages

Number of Text Messages (Bin)	Frequency
40-45	2
45-50	0
50-55	0
55-60	0
60-65	11
65-70	20
70-75	27
75-80	19
80-85	12
85-90	3
90-95	3
95-100	2
100-105	1
105-110	1

Seventh Graders

Frequency

Number of Text Messages

Number of Text Messages (Bin)	Frequency
40-45	0
45-50	0
50-55	0
55-60	0
60-65	0
65-70	0
70-75	7
75-80	31
80-85	32
85-90	25
90-95	4
95-100	0
100-105	0
105-110	0
110-115	0
115-120	0
120-125	0
125-130	1
130-135	0
135-140	0
140-145	0

- 



1. I can describe the shape of a histogram.

2. I can describe gaps, outliers, and clusters in a histogram.

3. I can describe the spread of a data set given a histogram.

Unit 9, Lesson 7: Describing and Interpreting Line Plots



Benchmark

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.



Learning Targets

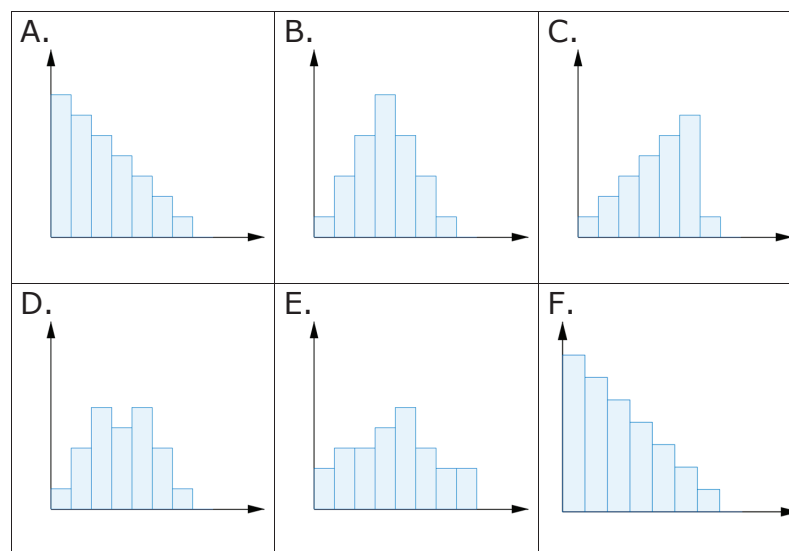
- ☐ I can describe and interpret data represented on a line plot.
- ☐ I can determine the similarities and differences between a line plot and a histogram.



9.7.1 Warm-Up: Gathering Data

1. Categorize each histogram according to its shape by placing the letter in the appropriate box in the chart below.

Symmetric	Skewed Left	Skewed Right





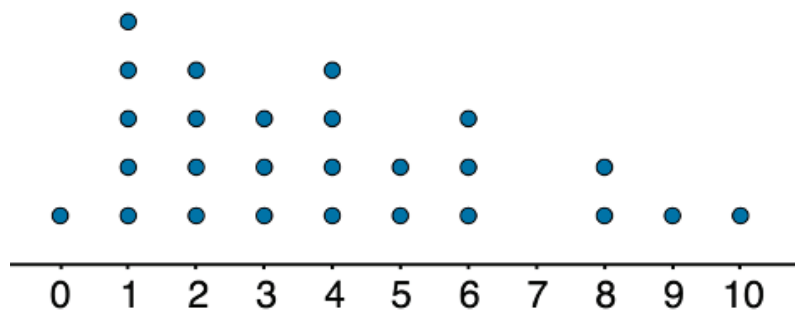
9.7.2 Guided Instruction: Describing and Interpreting Data Represented on Line Plots

Line plots are another way to display numerical data.



A **line plot** is a visual display of data values where each data value is shown as a dot or mark above a number line. This is also known as a dot plot.

1. The line plot shows the number of hours per week that students reported spending on homework.



Hours Spent on Homework per Week

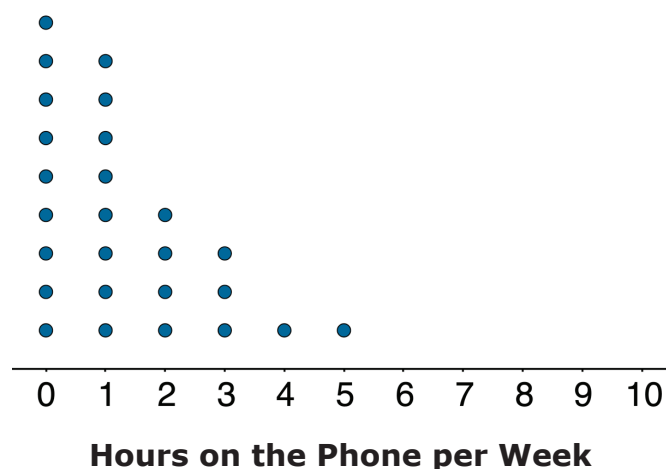
- a. How many students were surveyed?
- b. How many students reported spending 1 hour on homework each week?
- c. What percentage of the students reported spending 4 hours or fewer on homework each week?

- d. What is the range of time spent on homework in this group?

With your partner, discuss and summarize your discussion on the following questions about the hours spent on homework per week.

- e. Explain whether or not the mean time spent on homework by this group can be determined.
- f. Someone said, "In general, these students spend roughly the same number of hours doing homework." Do you and your partner agree? Explain your reasoning.
- g. Describe the shape and special features of the distribution of hours spent on homework. Find another pair of partners and compare your descriptions.

2. The same students were asked to estimate how many hours a week they spend talking on the phone. This line plot represents their reported number of hours of phone usage per week.

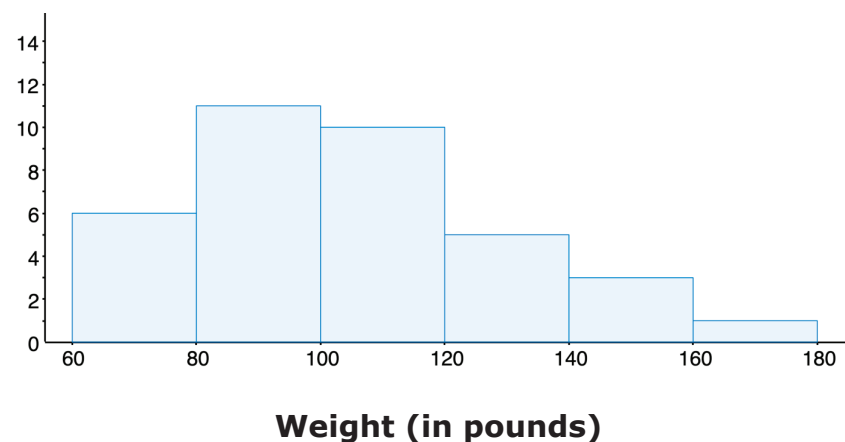
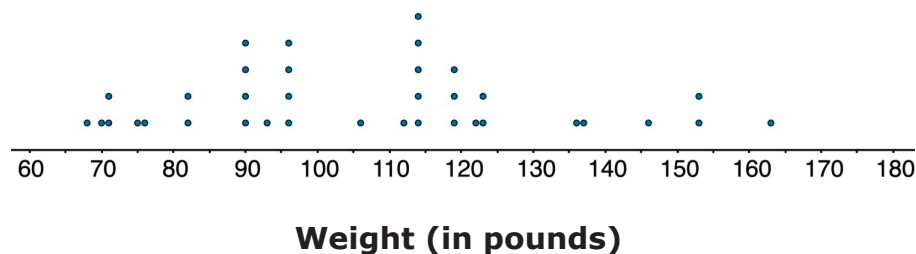


- Describe the spread of the data.
- Overall, are these students more alike in the amount of time they spend talking on the phone or in the amount of time they spend on homework? Explain your reasoning.
- Compare your reasoning with your partner. Make adjustments to your reasoning after the discussion with your partner, if necessary.



9.7.3 Exploration: Comparing Displays

1. The dot plot and histogram display the weights, in pounds, of 36 dogs at a dog show.



a.

- ☐ The line plot
- ☐ The histogram
- ☐ Both the line plot and histogram

can be used

to determine the number of dogs that weigh 90 lbs.

b. How many dogs have a weight of 90 lbs?

c. Find the range of the dogs' weights.

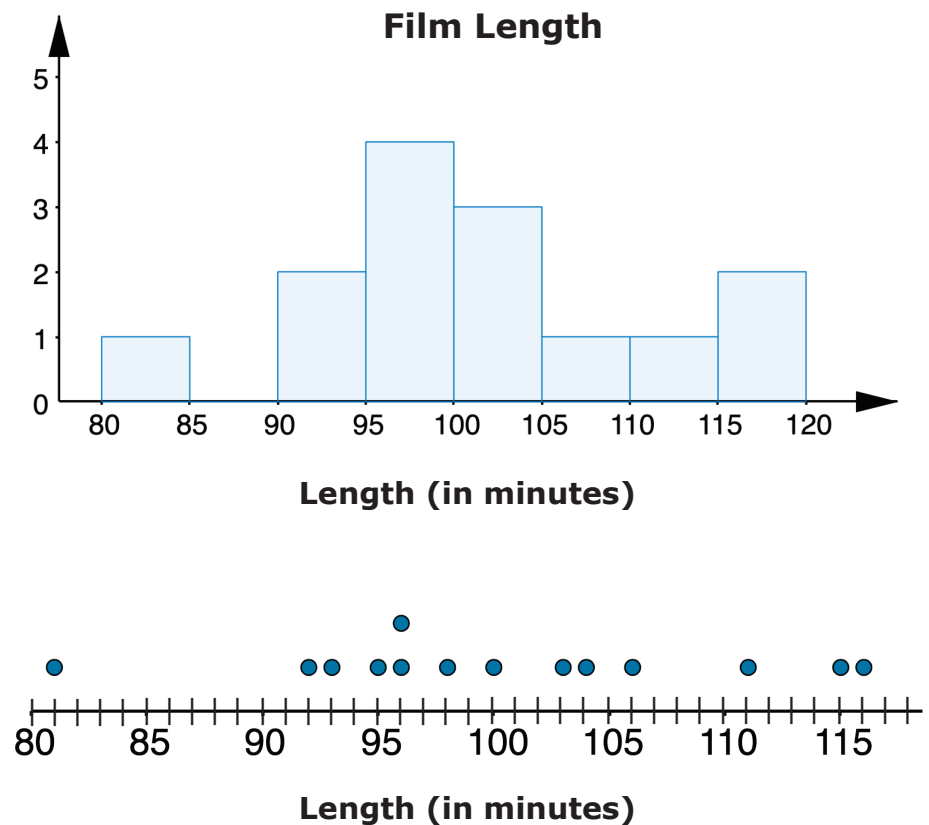
d. With your partner, discuss what you notice about the shape of the distribution.

e. Which display was more useful when describing the shape?



9.7.4 Your Turn!

1. A movie critic recorded the length, in minutes, of a sample of Pixar films and then made a histogram and a line plot of the data.



- a. How many film lengths are recorded?

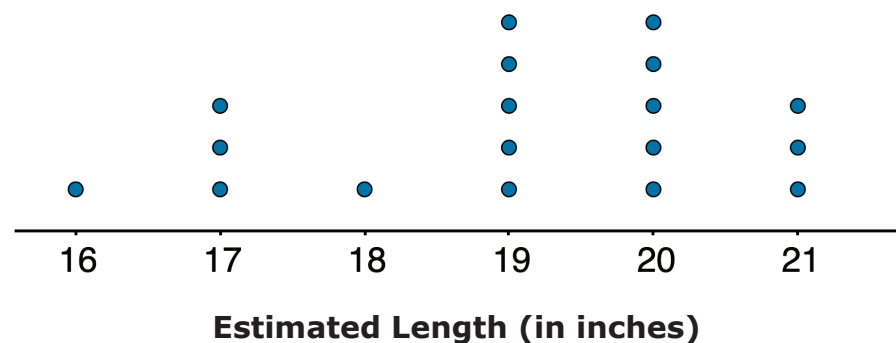
b. How many films are between 95 and 100 minutes in the histogram?

c. How many films are between 95 and 100 minutes in the line plot?

d. What is the value of the range?

e. Discuss with your partner which graph was used to determine the range and why.

2. A teacher drew a line segment that was 20 inches long on the blackboard. She asked each of her students to estimate the length of the segment and used their estimates to draw this line plot.



a. How many students were in the class?

b. Were students generally accurate in their estimates of the length of the line?

c. Describe the shape and distribution of the estimated lengths.



9.7.5 Wrap-Up: Reflection

1. Check all the statements that apply to your understanding.
 - ☐ I can read a line plot.
 - ☐ I need help reading a line plot.
 - ☐ I can describe and analyze data represented on a line plot.
 - ☐ I need help describing and analyzing data represented on a line plot.
 - ☐ I can determine the similarities and differences between a line plot and a histogram.
 - ☐ I need help to determine the similarities and differences between a line plot and a histogram for the same data.



Unit 9, Lesson 8: Comparing Distributions



Benchmarks

MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.



Learning Targets

- ☐ I can find and interpret the mean and median of data sets.
- ☐ I can use mean and median to compare data sets.
- ☐ I can compare the distribution of data sets.



9.8.1 Warm-Up: Which One Doesn't Belong?

- Which license plate doesn't belong? Explain your reasoning.



9.8.2 Collaboration: Comparing Distributions

- Clare recorded the amount of time students in the sixth, eighth, and tenth grades spent doing homework, in hours per week. She made a line plot of the data for each grade and provided the following summary.
 - Students in sixth grade tend to spend less time on homework than students in eighth and tenth grades.
 - The homework times for the tenth-grade students are more alike than the homework times for the eighth-grade students.

Use Clare's summary to match each line plot to the correct grade (sixth, eighth, or tenth).

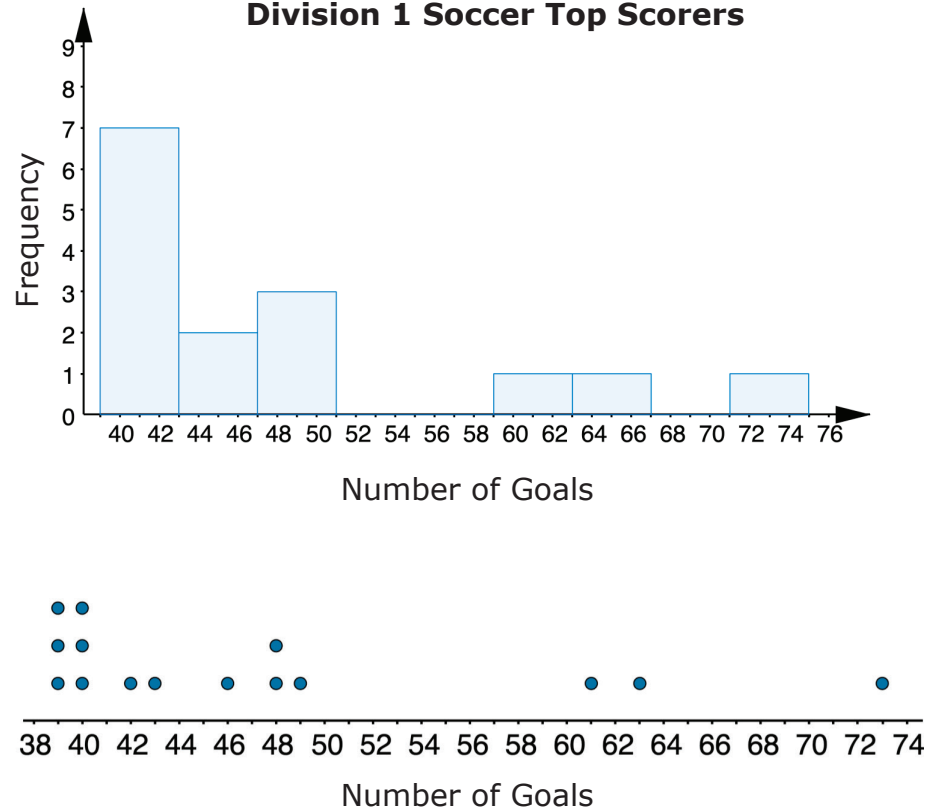
Grade Level	Line Plot



9.8.3 Guided Practice: Comparing Distributions

1. A histogram and line plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season are shown.

2019-2020 NCAA Women's Division 1 Soccer Top Scorers



- a. The

- ☐ histogram
- ☐ line plot
- ☐ histogram and line plot

can be used to

determine the mean.

- b. The

- ☐ histogram
- ☐ line plot
- ☐ histogram and line plot

can be used to

determine the median.

- c. Which measure of central tendency do you expect to be larger, the mean or the median? Explain your reasoning.

2. The table lists the data for the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.

Rank	Player	Goals
1	Evelyne Viens, South Fla.	73
2	Catarina Macario, Stanford	63
3	Raimee Sherle, Boise St.	61
4	Ally Watt, Texas A&M	49
5	Tziarra King, NC State	48
5	Deyna Castellanos, Florida St.	48
7	Alexa Genas, Campbell	46
8	Morgan Weaver, Washington St.	43
9	Peyton DePriest, Middle Tenn.	42
10	Elise Flake, BYU	40
10	Samantha Dewey, Xavier	40
10	Hayley Younginer, Wake Forest	40
13	Dakota Mills, Saint Joseph's	39
13	Katie McClure, Kansas	39
13	Taylor Kornieck, Colorado	39

- a. Find the mean number of goals. Interpret the mean.
- b. Find the median number of goals. Interpret the median.

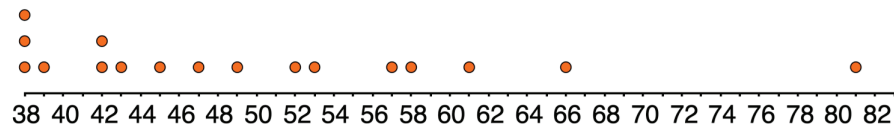
- c. Which value is larger, the mean or the median?
- d. How would the mean and median change if you found the mean goals and median goals for the top 8 scorers?
- e. Find the mean number of goals for the top 8 scorers.
- f. Find the median number of goals for the top 8 scorers.
- g. Discuss with your partner how the measures of central tendency changed. Write your conclusion.



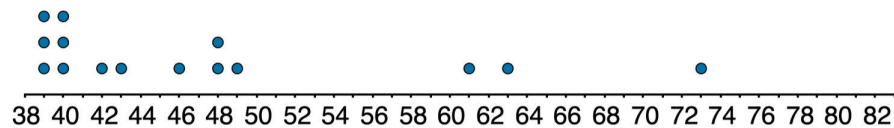
9.8.4 Your Turn!

1. The line plots show the top 15 women goal scorers during the 2016-2017 and 2019-2020 NCAA Women's Division 1 soccer season.

2016-2017 Season



2019-2020 Season



Number of Goals

- a. What is different about the top 15 goal scorers during the 2016-2017 soccer season and the top 15 goal scorers during the 2019-2020 soccer season?

- b. Make a hypothesis determining whether the mean of the 2019-2020 season or the mean of the 2016-2017 season is higher. Explain what influenced your hypothesis.
- c. Make a hypothesis determining whether the median of the 2019-2020 season or the median of the 2016-2017 season is higher. Explain what influenced your hypothesis.
- d. Compare your predictions with your partner's predictions. Make any adjustments needed based on your conversation.

2. The table shows the list of the top 15 women goal scorers during the 2016-2017 NCAA Women's Division 1 soccer season.

Rank	Player	Goals
1	Savannah Jordan, Florida	81
2	Tabitha Tindell, FGCU	66
3	Rachel Hill, UConn	61
4	Mckenzie Meehan, Boston College	58
5	Aaliyah Lewis, Alabama St.	57
6	Tyler Lussi, Princeton	53
7	Valerie Sanderson, Memphis	52
8	Murielle Tiernan, Virginia Tech	49
9	Ashley Hatch, BYU	47
10	Ariela Lewis, Alabama St.	45
11	Emily Gingrich, Saint Joseph's	43
12	Margaret Purce, Harvard	42
12	Jaycie Johnson, Nebraska	42
14	Hayley Dowd, Boston College	39
15	Rachel Holden, North Texas	38
15	Lauren Miller, North Dakota St.	38
15	Abby Reed, DePaul	38

- a. Find the mean number of goals for the 2016-2017 soccer season.

- b. Interpret the mean.

- c. Find the median number of goals for the 2016-2017 soccer season.

- d. Interpret the median.

- e. Complete the statements.

The mean of the 2016-2017 soccer season is

- ☐ less
☐ greater

than the mean of the 2019-2020 soccer

season. The value in the ☐ 2016-2017
☐ 2019-2020 soccer

season that likely influenced the mean is _____.

The median of the 2016-2017 soccer season is

- ☐ less
☐ greater

than the median of the 2019-2020

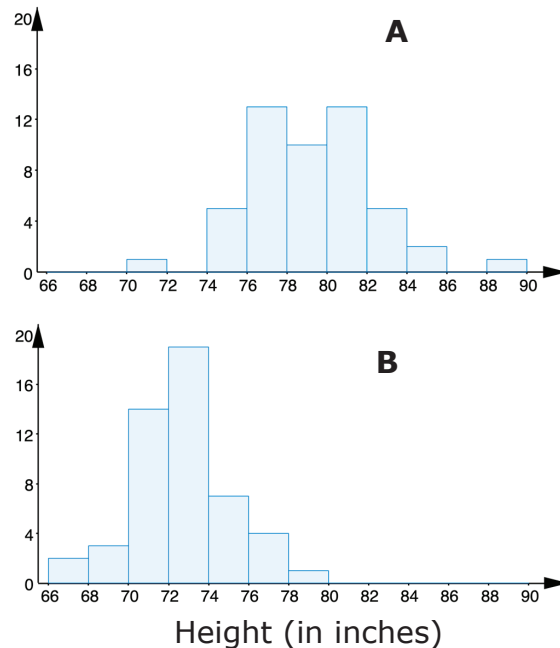
soccer season.



9.8.5 Wrap-Up: Ticket Out the Door

1. Professional basketball players tend to be taller than professional baseball players.

The two histograms, labeled A and B, show height distributions of 50 male professional baseball players and 50 male professional basketball players.



- a. Choose one of the titles: "Heights of Baseball Players" or "Heights of Basketball Players" for each histogram.

Histogram	A	B
Best Title		

- b. Describe the spread and shape of the distribution of the heights of the basketball players.

Unit 9, Lesson 9: Creating Box Plots



Benchmarks

MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data.

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.



Learning Targets

- ☐ I can find key values given a box plot within a real-world context.
- ☐ I can create a box plot to represent data.



9.9.1 Warm-Up: Sleep

Ten sixth-grade students were asked how much sleep, in hours, they usually get on a school night. Their responses are shown.

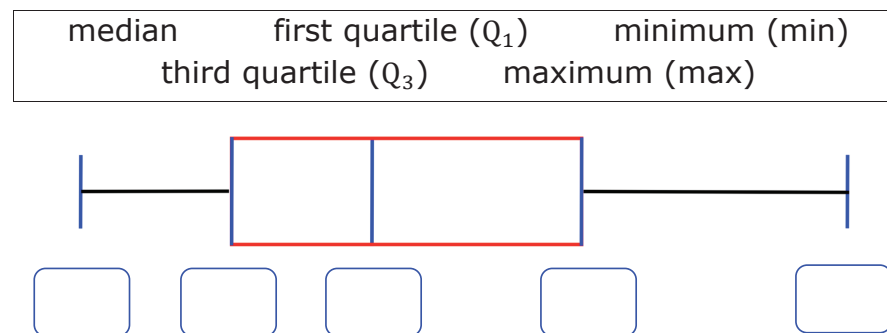
6, 7, 6.5, 7.5, 8, 8.5, 7, 6.5, 5, 7

1. Calculate the median number of hours of sleep for the students.
2. Calculate the range for number of hours of sleep for the students.



9.9.2 Refresher: Box Plots

A blank box plot is shown. Label the blue boxes below using the words found in the bank.



1. Complete the statements.

The _____ is a measure of spread determined by finding the distance between the first quartile and the third quartile. It measures the spread of the middle _____ of the data.

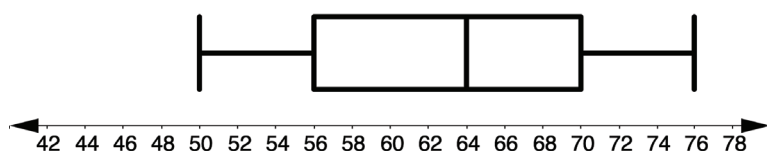
The _____ is another measure of spread determined by finding the difference between the largest data value and the smallest data value.

2. Complete each statement.

- a. _____% of the data falls between the minimum and the first quartile (Q_1).
- b. _____% of the data falls between the minimum and the third quartile (Q_3).
- c. _____% of the data falls between the first quartile (Q_1) and the third quartile (Q_3).

3. The box plot shows the distribution of the daily number of people who were at a large community pool in July 2018.

Number of Patrons at Community Pool



- What is the median number of people at the pool each day?
- What is the maximum number of people?
- What is the minimum number of people?
- Determine the value of the first quartile (Q_1) from the box plot.
- Determine the value of the third quartile (Q_3) from the box plot.



The **five-number summary** are the values of the minimum, first quartile (Q_1), median, third quartile (Q_3), and the maximum. These are the values used to create a box plot to represent a data set.



9.9.3 Guided Instruction: The Five-Number Summary

- The ages of 20 customers who attended a special event at a local restaurant are shown in order from least to greatest.

7, 8, 9, 10, 10, 11, 12, 15, 16, 20, 20,
22, 23, 24, 28, 30, 33, 35, 38, 42

- Find the five-number summary of the data set.

Minimum: _____

First quartile: _____

Median: _____

Third quartile: _____

Maximum: _____

- Discuss with your partner if the data the restaurant collected is reflective of the ages of customers who ate at the restaurant.



Truthful data is data that is random, representative of the population, and has a large sample size.

2. The birth weights, in ounces, of golden retriever puppies born at a kennel were recorded for the past month. The data is recorded in the table.



13	14	15	15	16	16	16	16	17
17	17	17	17	17	17	18	18	18
18	18	18	18	18	19	20		

- a. Find the five-number summary of the data set.

Minimum: _____

First quartile: _____

Median: _____

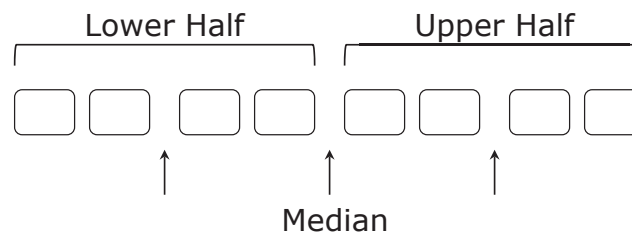
Third quartile: _____

Maximum: _____

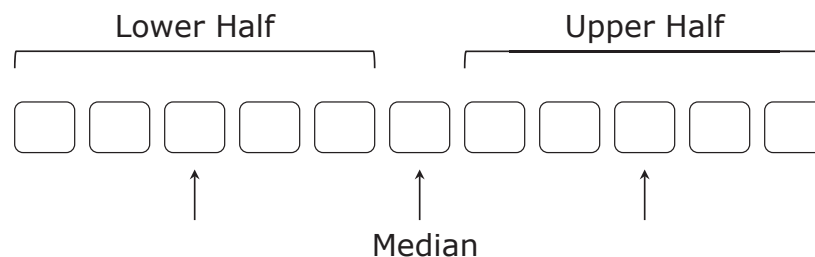
- b. Discuss with your partner why the data may not be a true representation of the birth weight of all puppies.

3. What differences did you notice when calculating the five-number summary for the two different data sets?

An illustration of the five-number summary for an even data set is shown.



An illustration of the five-number summary for an odd data set is shown.



4. Discuss with your partner your observations and any questions you have about the illustrations.



9.9.4 Guided Practice: Creating a Box Plot

- Using the data and five-number summary on the birth weight, in ounces, of the golden retriever puppies born at a kennel in the last month, create a box plot from the steps listed.

13	14	15	15	16	16	16	16	17
17	17	17	17	17	17	18	18	18
18	18	18	18	18	19	20		

Step 1: Label the axis by starting at a number slightly less than the minimum. Increase by the same interval, making sure each interval is equally spaced, until you reach (or pass) the maximum.



Step 2: Place a dot above the axis for each value of the five-number summary.

Step 3: Draw the box from Q_1 to Q_3 with a line inside the box to represent the median.

Step 4: Connect the minimum and the maximum to each side of the box with a line.

Step 5: Include a title and a label for the axis.

- Compare your box plots with your partner. Make any revisions if necessary.



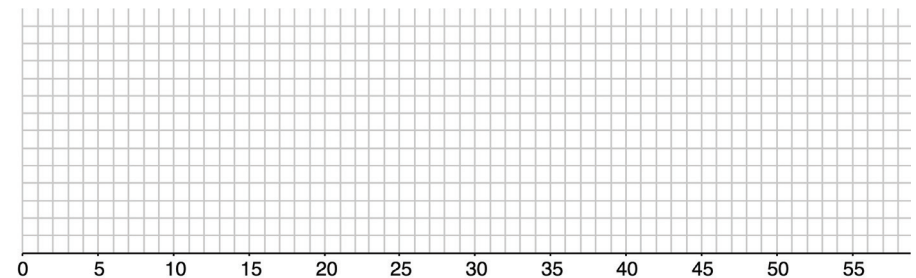
9.9.5 Try It: Creating a Box Plot

- The ages of the 21 restaurant customers who attended a special event at a local restaurant are shown.

27, 28, 39, 18, 18, 21, 32, 25, 46, 40, 40,
22, 23, 24, 28, 30, 33, 35, 38, 42, 57

- Determine the five-number summary for the data set.

- Create a box plot using the data and five-number summary for the 21 restaurant customers.



- Does the length of each part of the box plot need to be the same?

d. Why is the actual length of each part not equal, even though they each represent 25% of the data?

e. What is the interquartile range for the data?

f. What is the range?



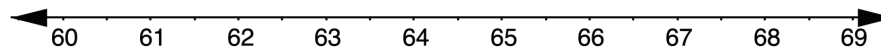
9.9.6 Your Turn!

1. Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. The five-number summary for his pass-completion percentages from 2000 to 2009 is shown.

Minimum	Maximum	Q_1	Q_3	Median
60.2	68.9	61.8	65.7	63.3

- a. Discuss with your partner why the data used to determine the five-number summary for Tom Brady's pass completion is more truthful than the data for golden retriever puppies, or the ages of the restaurant customers who attended a special event.
- b. Use the five-number summary to draw and label a box plot of the data set.

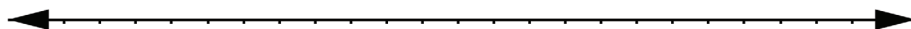
Tom Brady's Completion Percentages 2000-2009



- c. Determine an alternative scale that could be used to make the box plot.

- d. What scale would you use if the data values ranged from 10.3 to 48.4?

- e. Label the number line with your scale from Part D. Then, compare it with your partner.

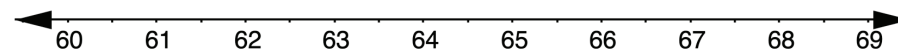


2. Tom Brady's pass-completion percentages from 2009 to 2019 are listed in the table.

Year	2009	2010	2011	2012	2013	2014
%	65.7	65.9	65.6	63.0	60.5	64.1
Year	2015	2016	2017	2018	2019	
%	64.4	67.4	66.3	65.8	60.8	

- a. Create a box plot of Tom Brady's pass-completion percentages.

Tom Brady's Pass-Completion Percentages 2009-2019



- b. Determine the interquartile range for Tom Brady's pass-completion percentages.

- c. What is the range of pass-completion percentages?



9.9.7 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of creating box plots. Then provide a brief explanation or evidence to support your self-assessment.

1. I can find the five-number summary of a data set.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



2. I can create an axis with the appropriate scale.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



3. I can find and interpret the interquartile range.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



Unit 9, Lesson 10: Creating Histograms



Benchmark

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.



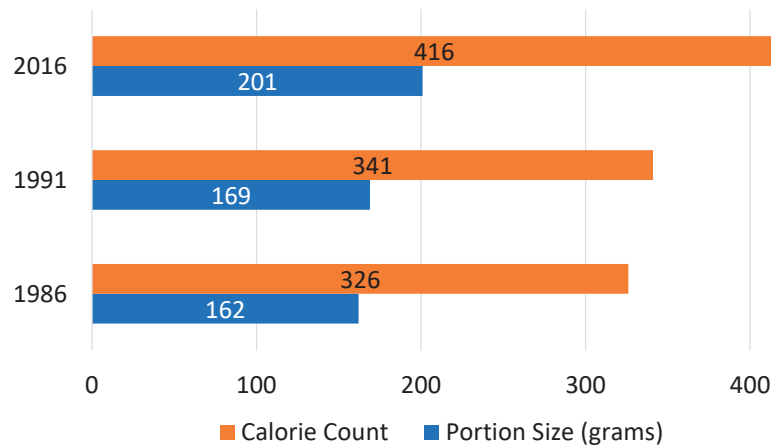
Learning Target

- ☐ I can create a histogram to represent data.



9.10.1 Warm-Up: Notice and Wonder

The portion size and calorie count of fast-food menu items have changed over time.



1. What are two things you notice about the graph?

2. What is one thing you wonder about the graph?



9.10.2 Guided Instruction: Elements of Creating Histograms

- An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. The farmer measured the lengths of 24 earthworms in one of the containers. Each length was measured in millimeters and recorded in the table.

6.5	11.2	18	19.6	20.1	23.8	23.5	25
25.7	26.2	27.5	27.5	28	29.9	32	33
41.4	42.3	48.8	52	54.4	59	60.6	77

Work with your partner to collect the pieces needed to create a histogram.

- The range of the data is _____.

Determine the **interval width** by dividing the range and the number of intervals desired.

$$\frac{\text{Range}}{\text{Number of Intervals}} \approx \text{Interval Width}$$

- Discuss three different numbers of intervals you could use with your partner, and how using a different number of intervals would affect the histogram.
- For this histogram, use six intervals. Divide the range by _____ and round up to the nearest whole number. This gives an **interval width** of _____.



A **frequency table** is a table that shows how often each item, number, or range of numbers occurs in a set of data.

Use the steps below to create a frequency table to represent the data.

- d. Complete each statement.

The minimum value is _____, so create the first interval that starts at 6 (or less).
To get the highest value in the interval, add the interval width of 12.

The first interval is 6 – 18, the second interval is 18 – _____.

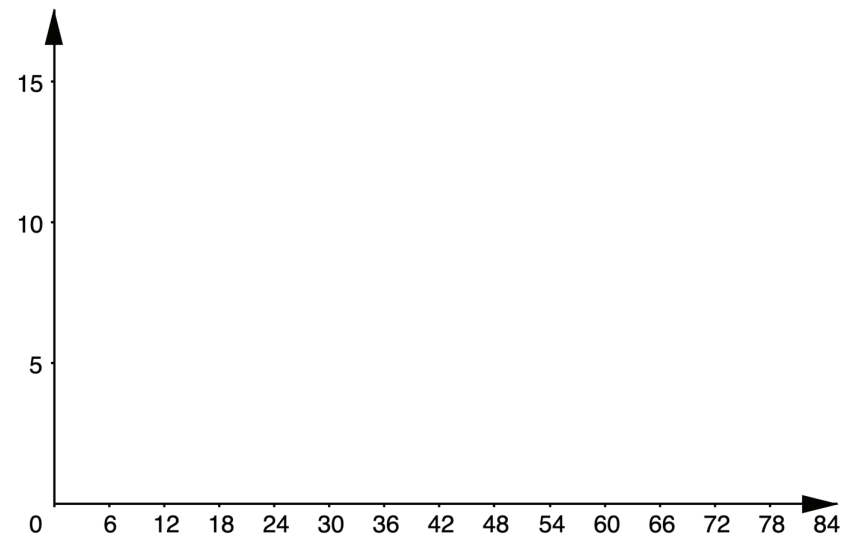
- e. With your partner, discuss if the value of 18 can be included in both intervals. Explain your conclusion.
- f. Complete the **Length (mm)** column of the table shown to indicate the intervals.

Length (mm)	Frequency
6 – 18	
18 –	

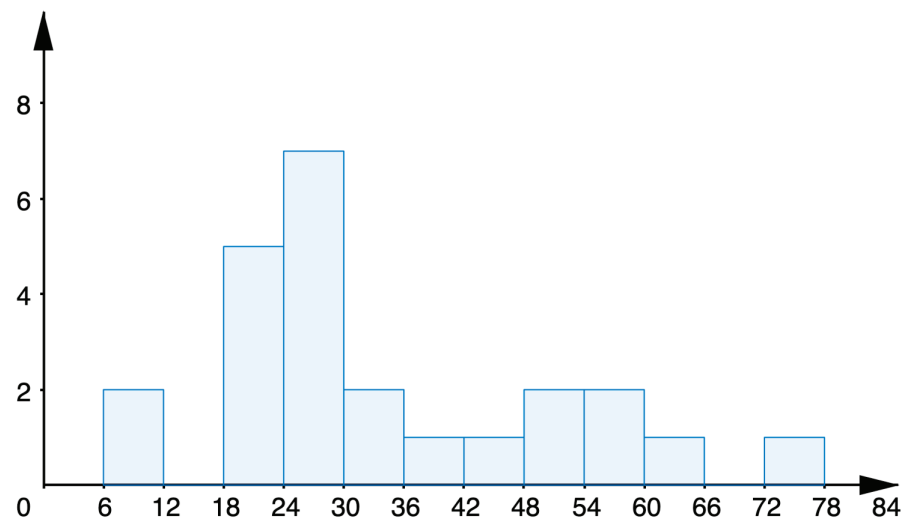
- g. With your partner, determine the frequency values and fill in the **Frequency** column of the table.

In a histogram, the bars touch. The variable measured is placed on the horizontal axis, and frequency is placed on the vertical axis.

- h. Label the horizontal and vertical axis of the grid below. Then, use the grid to create the histogram to display the lengths of earthworms in millimeters.



2. The same data could be displayed in this histogram.



- With your partner, discuss the differences you notice about this histogram from the one you created.
- Explain which histogram better displays the data.

c. Ask two classmates the reasons they listed for which histogram better displays the data. Record their choice and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Histogram Choice	My Summary of their Reason	Initials

d. Explain whether the data has the characteristics that will result in truthful data.



9.10.3 Guided Practice: Creating Histograms

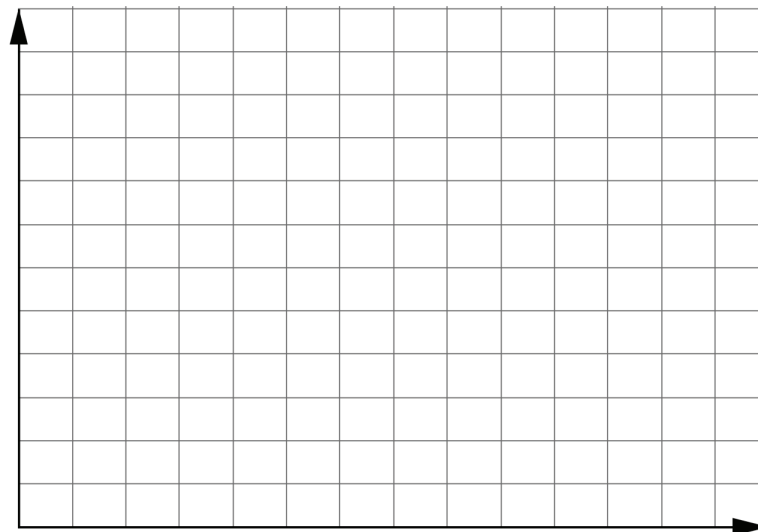
1. The data shows the number of Snapchats sent by 12 seventh-grade students in one day.

99, 97, 94, 88, 84, 81, 80, 77, 71, 63, 60, 54

Complete the following to create a frequency table and a histogram to represent the number of Snapchats sent by seventh-grade students in one day.

- How many intervals are appropriate to use for this histogram?
- List three possible interval widths.
- Using one of the possible interval widths, create the frequency table and histogram of the data. Make sure to label both the horizontal and vertical axis of the histogram.

Snapchats	Frequency



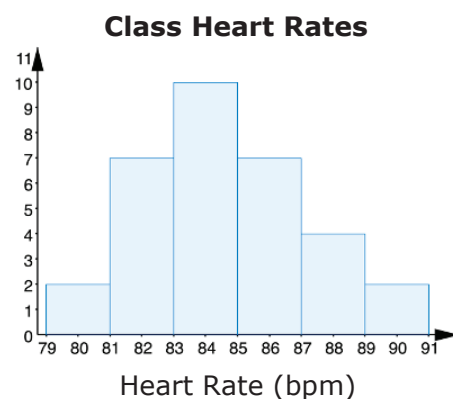
- Find a classmate who used a different interval width and compare your histograms. Summarize your discussion on what you notice about the two different graphs.

2. Raisa used data she collected about the resting heart rates, in beats per minute (bpm), of 22 classmates to create a histogram. The heart rates are listed below.

89 87 85 84 90 79 83 85 86 88 84 81 88 85 83
83 86 82 83 86 82 84

She created a frequency table and used it to make a histogram. The frequency table and histogram Raisa created are shown below. Unfortunately, Raisa noticed the data she collected was not accurately displayed.

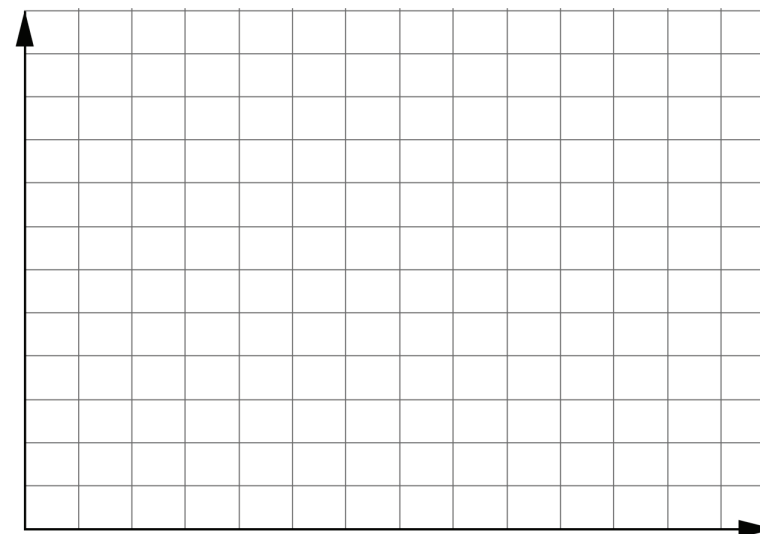
Heart Rate (bpm)	Frequency
79 – 81	2
81 – 83	7
83 – 85	10
85 – 87	7
87 – 89	4
89 – 91	2



- a. Explain what Raisa did wrong.

- b. Correct Raisa's mistake and create a frequency table and histogram that correctly displays the data she collected.

Heart Rate (bpm)	Frequency





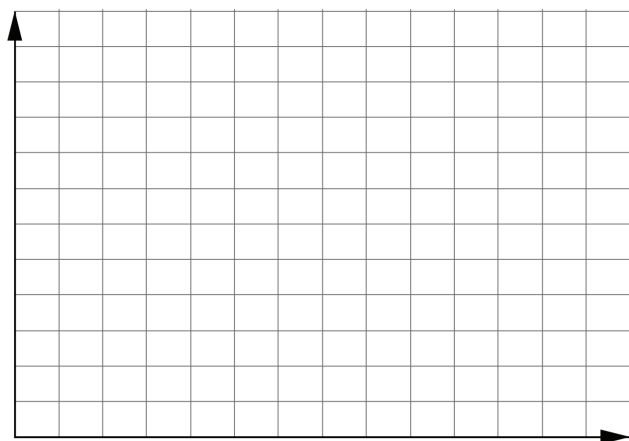
9.10.4 Your Turn!

1. Sergio noticed that a bag of chips advertised that the serving size was 15 chips and that the bag contained approximately 4 servings. He counted the number of chips in a sample of 12 bags and recorded the total number of chips below.

48, 51, 52, 55, 56, 58, 59, 60, 61, 62, 65, 67

Create a frequency table and histogram for the number of chips in a bag.

Number of Chips	Frequency



9.10.5 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on creating histograms?

